

A low-angle, upward-looking perspective of several modern skyscrapers with glass facades. The buildings are arranged in a way that they appear to converge towards the top of the frame, creating a sense of height and scale. The sky is a clear blue with some wispy white clouds. The text is overlaid on the upper portion of the image.

*Are TN counties becoming more
vulnerable to extreme weather over
time?*

By Gabriel Hardin

Purpose

- My goal is to demonstrate the unique vulnerabilities to storms of different TN counties to inform government, policy makers, and county residents.
- “How much in damages do TN counties experience because of storms, and how has that changed over time.”
- Using Data from the NOAA Storm Events database, I rank ordered the severity of disasters and broke them down by county and disaster type.



Methodology

1

Gathering data
from NOAA
Storm Events
Database

2

Joining and
cleaning data
from CSVs in
Python

3

Using Regex to
extract damage
metrics from text
data

4

Creating and
modeling data
tables in SQL

5

Visualizing data
in Power Bi

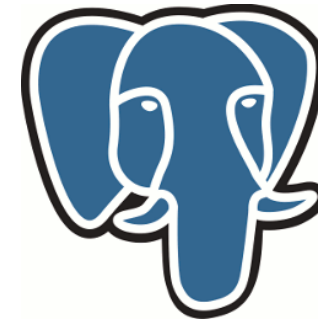
Defining “damage severity”

- Using Regex, I analyzed incident reports from text data which contained words like “power loss”, “power outages”, “road closure”, and “road closed”.
- This determined which storm events did or did not result in road closures and power outages.
- To create a “damage severity” score, I created a percentile score ranking each disaster report from 0.0 – 1.0 using data columns for road closures, power outages, fatalities, injuries, property damages, and crop damages.
- C:\Users\plann\Documents\Nashville Software School\Python\Capstone\TN_storms\python_cleaning\regex_narratives.ipynb



Data Modeling

- Using the CREATE and DROP commands in PostgreSQL, I imported csv data into pgAdmin 4.
- First, I imported flat csv files with data relevant to my dashboard, then I divided the large flat file into a model called a “Star Schema”
- C:\Users\plann\Documents\Nashville Software School\Python\Capstone\TN_storms\sql_analysis_modelling\star_schema.sql



PostgreSQL
the world's most advanced open source database

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The Dashboard

- C:\Users\plann\Documents\Nashville Software School\Python\Capstone\TN_storms\sql_analysis_modelling\Gabe_hardin_NSS_capstone.pbix